

What is AXI:

- Stand for Advanced eXtensible Interface
- It is part of ARM's **AMBA** (Advanced Microcontroller Bus Architecture) family.

AXI Variants

AXI4 (Full AXI)

This is the complete protocol.

Supports:

- Read
- Write
- Burst transfers
- Large data widths
- High performance

Typical use:

- CPU ↔ DDR
- DMA ↔ Memory
- GPU ↔ Memory

AXI4-Lite

This is a simplified version.

Supports:

- Single read
- Single write
- No bursts

Perfect for:

- GPIO
- UART
- SPI
- Timers
- Control Registers

AXI4-Stream

Completely different.

- No addresses.
- Only streaming data.

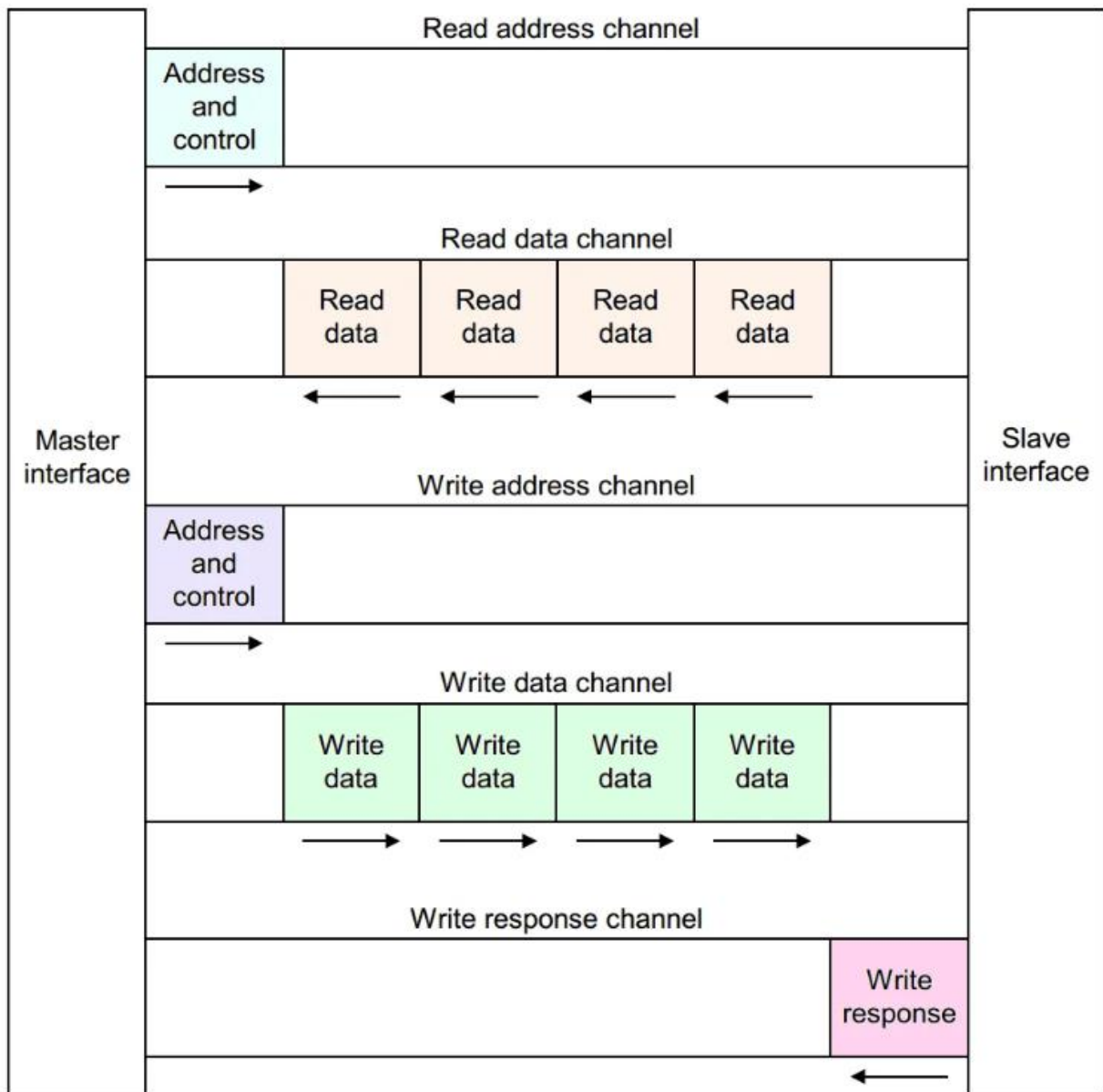
Example:

- Video
- Audio
- Network Packets
- AI Accelerator

AXI Channels

There are five independent channels between an AXI master and slave. They are the:

1. Read address channel
2. Read data channel
3. Write address channel
4. Write data channel
5. Write response channel



AXI Channels Signals

Read Address Channel Signals

ARID[3:0] Read address ID	Master
ARADDR[31:0] Read address	Master
ARLEN[3:0] Burst length	Master
ARSIZE[2:0] Burst size	Master
ARBURST[1:0] Burst type	Master
ARLOCK[1:0] Lock type	Master
ARCACHE[3:0] Cache type	Master
ARPROT[2:0] Protection type	Master
ARVALID Read address valid	Master
ARREADY Read address ready	Slave

Global Signals

ACLK Global clock signal	Clock source
ARESETn Global reset signal	Reset source

Read Data Channel Signals

RID[3:0] Read ID tag	Slave
RDATA[31:0] Read data	Slave
RRESP[1:0] Read response	Slave
RLAST Read last	Slave
RVALID Read valid	Slave
RREADY Read ready	Master

Write Address Channel Signals

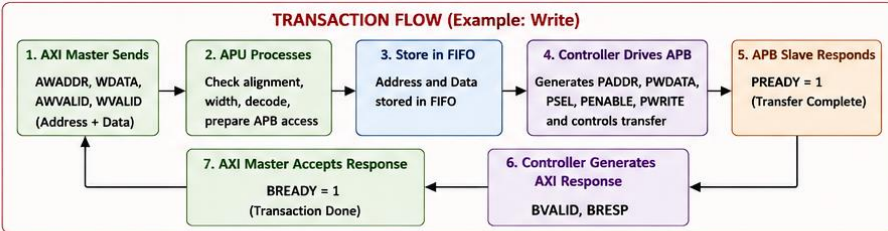
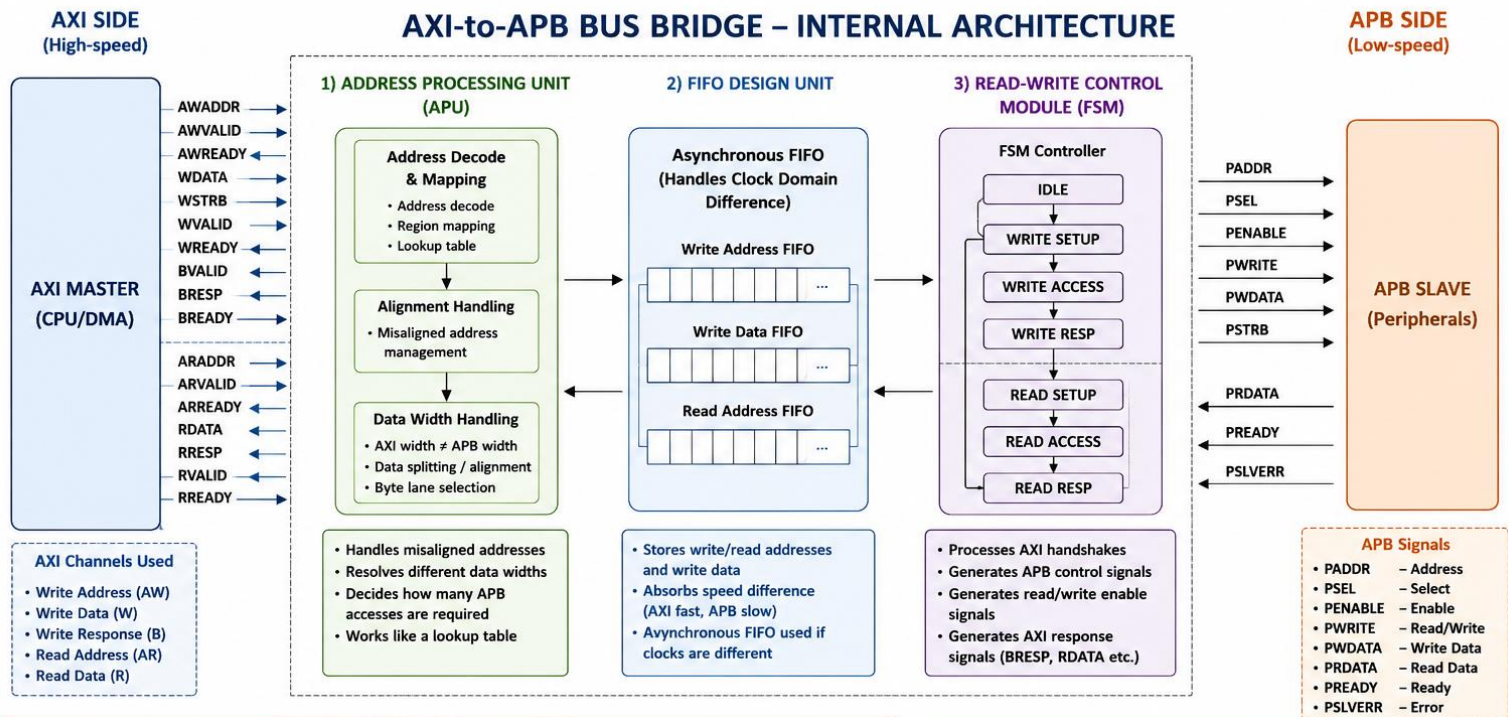
AWID[3:0] Write address ID	Master
AWADDR[31:0] Write address	Master
AWLEN[3:0] Burst length	Master
AWSIZE[2:0] Burst size	Master
AWBURST[1:0] Burst type	Master
AWLOCK[1:0] Lock type	Master
AWCACHE[3:0] Cache type	Master
AWPROT[2:0] Protection type	Master
AWVALID Write address valid	Master
AWREADY Write address ready	Slave

Write Data Channel Signals

WID[3:0] Write ID tag	Master
WDATA[31:0] Write data	Master
WSTRB[3:0] Write strobes	Master
WLAST Write last	Master
WVALID Write valid	Master
WREADY Write ready	Slave

Write Response Channel Signals

BID[3:0] Response ID	Slave
BRESP[1:0] Write response	Slave
BVALID Write response valid	Slave
BREADY Response ready	Master

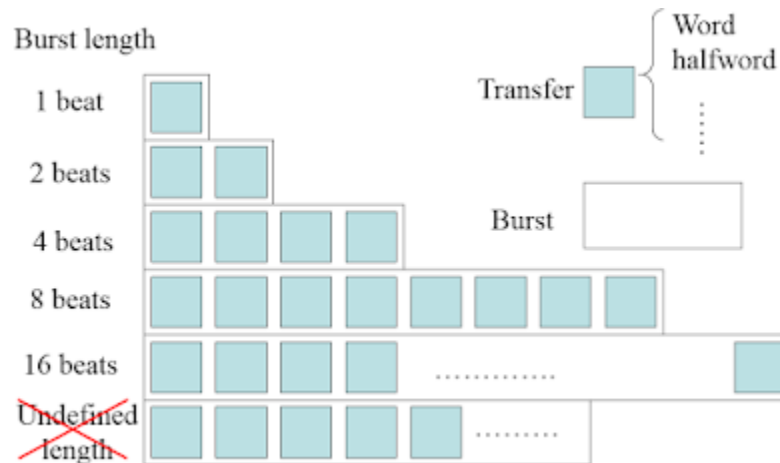


AXI is high-performance and can send bursts of data. APB is simple and slow (no burst).
The bridge collects AXI requests, stores them, and converts them into single APB transfers.

Burst operation:

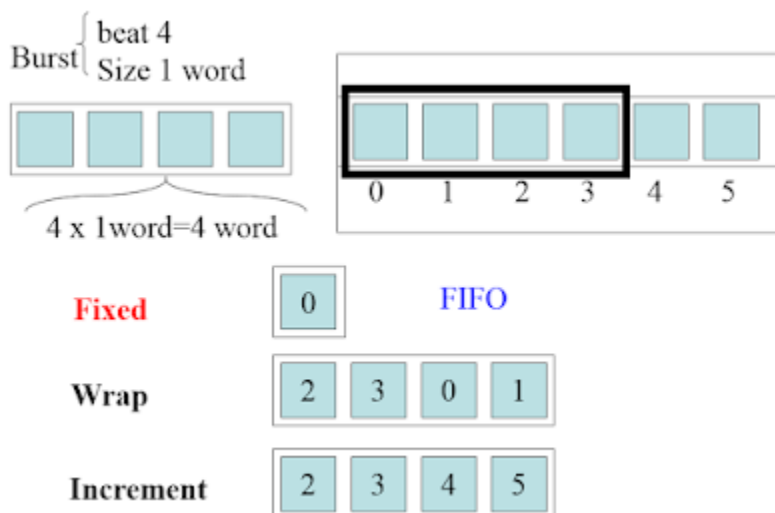
In AXI channel, the number of data transfers in a single burst are called as beats.

A burst is a group of consecutive beats.

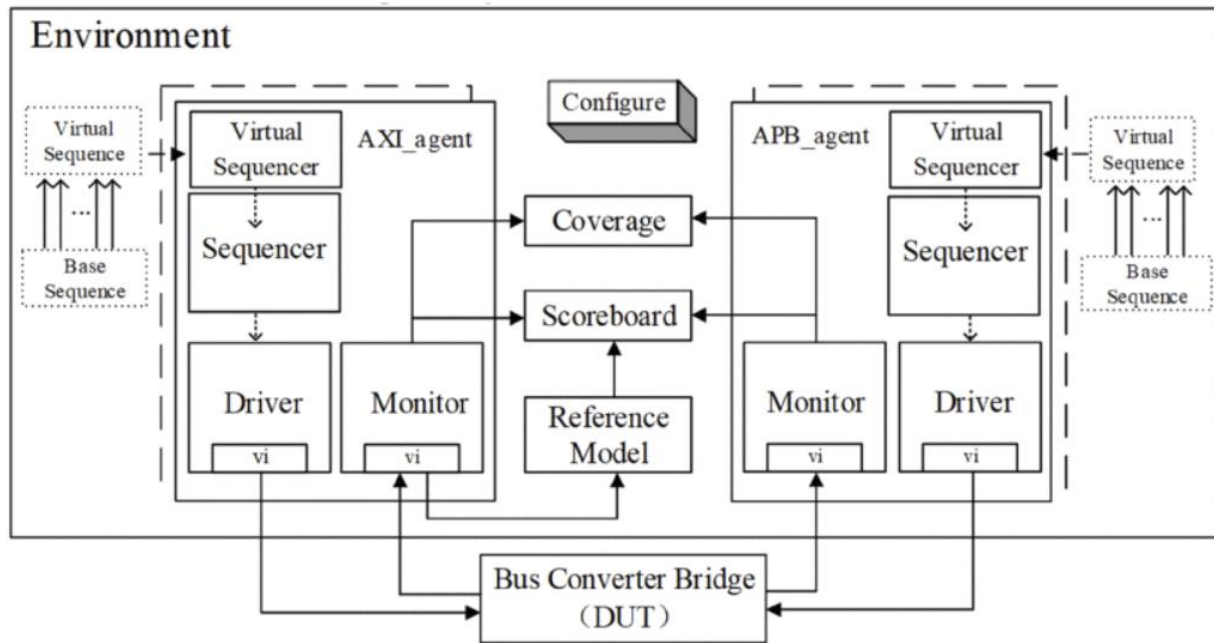


Burst Transfer Type:

1. **Fixed burst:** In a fixed burst, the address remains the same for every transfer in the burst. This burst type is for repeated accesses to the same location such as when loading or emptying a peripheral FIFO.
2. **Incrementing burst:** In an incrementing burst, the address for each transfer in the burst is an increment of the previous transfer address. The increment value depends on the size of the transfer. For example, the address for each transfer in a burst with a size of four bytes is the previous address plus four.



3. **Wrapping burst:** A wrapping burst is similar to an incrementing burst, in that the address for each transfer in the burst is an increment of the previous transfer address. However, in a wrapping burst the address wraps around to a lower address when a wrap boundary is reached. The wrap boundary is the size of each transfer in the burst multiplied by the total number of transfers in the burst. There are two restrictions for wrapping bursts: the start address must be aligned to the size of the transfer and the length of the burst must be 2, 4, 8, or 16.



Test Plan

Test Name	AXI Sequence	APB Slave Behavior	What is Verified
Write Test	Single Write	Normal	AXI write → APB write
Read Test	Single Read	Return data	APB read → AXI read
Random Test	Random R/W	Normal	General functionality
Burst Test	Burst Write/Read	Normal	Burst conversion (Full AXI only)
Wait-State Test	Read/Write	Delayed PREADY	Bridge waits correctly
Error Test	Read/Write	PSLVERR=1	Error propagation
Reset Test	Transactions after reset	Normal	Reset recovery
Stress Test	Thousands of random transactions	Random delays/errors	Robustness